

1. (X, Y) def. prin $f_{(X,Y)} = 2 \cdot e^{-x} \cdot e^{-2y} \cdot 1_{(0,\infty)}(x) \cdot 1_{(0,\infty)}(y)$

Verif. dacă X și Y sunt indep.

$$\Gamma(a) = \int_0^{\infty} x^{a-1} \cdot e^{-x} dx$$

$$f(x, y) = f_x(x) \cdot f_y(y)$$

$$f_x(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^{\infty} 2 \cdot e^{-x} \cdot e^{-2y} dy = 2 \cdot e^{-x} \cdot \underbrace{\int_0^{\infty} e^{-2y} dy}_{I_1} \Rightarrow f_x(x) = 2 \cdot e^{-x} \cdot \frac{1}{2} = e^{-x}, \text{ pentru } x \in (0, \infty)$$

s.v. $2y = t \Rightarrow y = \frac{t}{2} \Rightarrow dy = \frac{1}{2} dt$

$y=0 \Rightarrow t=0$

$y \rightarrow \infty \Rightarrow t \rightarrow \infty$

$$I_1 = \int_0^{\infty} e^{-t} \cdot \frac{1}{2} dt = \frac{1}{2} \cdot \underbrace{\int_0^{\infty} t^{1-1} \cdot e^{-t} dt}_{\Gamma(1)} = \frac{1}{2} \cdot \overset{1}{\Gamma(1)} = \frac{1}{2}$$

$$f_y(y) = \int_{-\infty}^{\infty} f(x, y) dx = \int_0^{\infty} 2 \cdot e^{-x} \cdot e^{-2y} dx = 2 \cdot e^{-2y} \int_0^{\infty} e^{-x} dx = 2 \cdot e^{-2y} \int_0^{\infty} (-e^{-x})' dx = 2 \cdot e^{-2y} \cdot (-e^{-x}) \Big|_0^{\infty} = 2 \cdot e^{-2y} \cdot (0 + 1) = 2 e^{-2y} \text{ pentru } \underline{\underline{y \in (0, \infty)}}$$

$$f_x(x) \cdot f_y(y) = e^{-x} \cdot 2 \cdot e^{-2y} \cdot 1_{(0,\infty)}(x) \cdot 1_{(0,\infty)}(y) = f(x, y) \Rightarrow \underline{\underline{X, Y \text{ indep.}}}$$